ESSENTIAL-SPECTRUM OBSTRUCTION FOR FINITE-SHIFT DILATION RESTRICTIONS IN HILBERT–POLYA REALIZATIONS

ARGUS OPEN RESEARCH PROJECT

Abstract. A recurring approach to the Hilbert–Pólya conjecture restricts the self-adjoint dilation generator $D = -i(x\frac{d}{dx} + \frac{1}{2})$ on $L^2((0,\infty), dx)$ to subspaces annihilating the trivial pole of the Riemann zeta function via shift operators $(S_a f)(x) = f(x-a)\mathbf{1}_{(a,\infty)}(x)$. We prove that for every nonzero complex polynomial $P(z)$ satisfying $P(1) = 0$ and having no roots in the open unit disk $|z| < 1$, the restriction of D to the common dense domain $P(S)C_c^\infty(0,\infty)$ admits singular Weyl sequences at every real spectral parameter $\lambda \in \mathbb{R}$ with explicit residual decay $O(1/n)$. Consequently, every self-adjoint extension of this restriction—both in $L^2(0,\infty)$ and in the graph-norm completion $\|P(S)f\|_2$—has essential spectrum equal to the entire real line $\mathbb{R}$, ruling out a purely discrete zero-ordinate spectrum for this class of realizations.

1. Main Mathematical Results

Theorem 1.1 (Full-line essential spectrum for finite-shift dilation restrictions)**.** *Let S be the unit right shift on $L^2((0,\infty), dx)$ and $D = -i(x\frac{d}{dx} + \frac{1}{2})$ the self-adjoint dilation generator. For every nonzero polynomial $P \in \mathbb{C}[z]$ with $P(1) = 0$ and $P(z) \neq 0$ for all $|z| < 1$, every self-adjoint extension $\widetilde{D}$ of $D|_{P(S)C_c^\infty(0,\infty)}$ satisfies*

$$\sigma_{\mathrm{ess}}(\widetilde{D}) \;=\; \mathbb{R}.$$

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1275 in the domain of spectral theory and analytic number theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* For every nonzero complex polynomial P with P(1)=0 and no root in the open unit disk, every self-adjoint extension of the dilation generator restricted to P(S)C_c^infinity(0,infinity) has essential spectrum equal to R.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- General Hilbert-Polya spectral realizations and the Riemann Hypothesis

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1275 (spectral theory and analytic number theory), archived in the accompanying repository package `results/1275/`.

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